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Editorial Board
Journal of the ACM
Association for Computing Machinery

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

"On the Thermodynamic Separation of P and NP: Computational Irreversibility and the Landauer Barrier,"

for consideration for publication in the *Journal of the ACM*.

Summary. The paper introduces a physically grounded framework for separating P from NP via Landauer's principle and the thermodynamic arrow of time. The central contribution is the *Thermodynamic Church-Turing Thesis* (TCTT)—the axiom that all physical computers obey Landauer's bound—under which $\mathsf{P} \neq \mathsf{NP}$ follows as a consequence of the Second Law of Thermodynamics.

Contributions. The paper provides:

- A formal *thermodynamic complexity measure* $\Theta(f, n)$ quantifying the irreducible entropy cost of computing f.
- A proof that one-way functions exhibit a super-polynomial asymmetry between evaluation and inversion thermodynamic costs.
- A conditional separation $\mathsf{P} \neq \mathsf{NP}$ under the TCTT.
- Explicit evasion of all three known barriers: relativization (Baker-Gill-Solovay), natural proofs (Razborov-Rudich), and algebrization (Aaronson-Wigderson).

Suggested Reviewers.

1. **Prof. Scott Aaronson** (UT Austin) — World authority on complexity theory and the physics of computation.
2. **Prof. Russell Impagliazzo** (UC San Diego) — Author of the "five worlds" framework for P vs NP.
3. **Prof. Avi Wigderson** (IAS Princeton) — Co-author of the algebrization barrier and expert in derandomization.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Respectfully submitted,

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